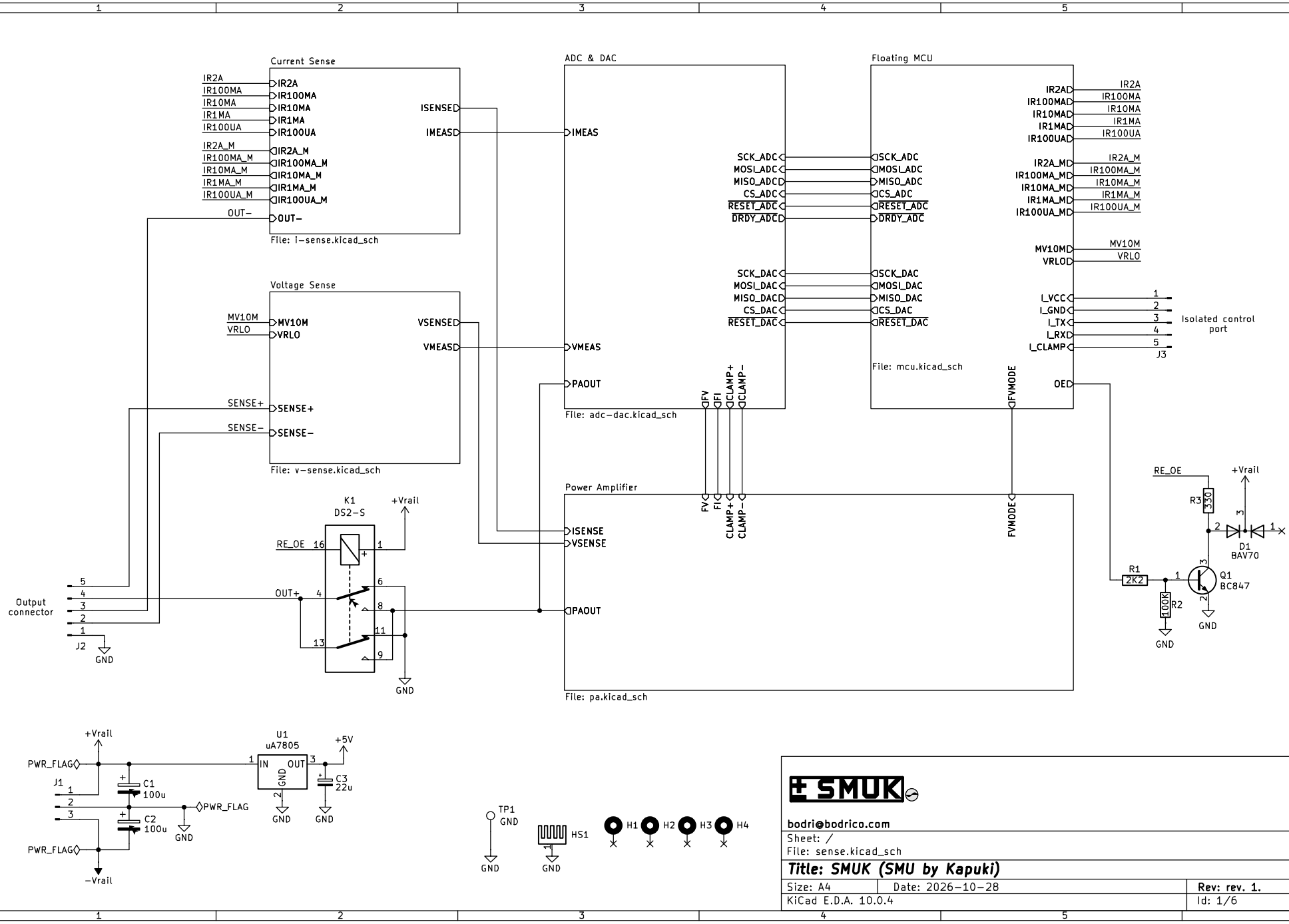
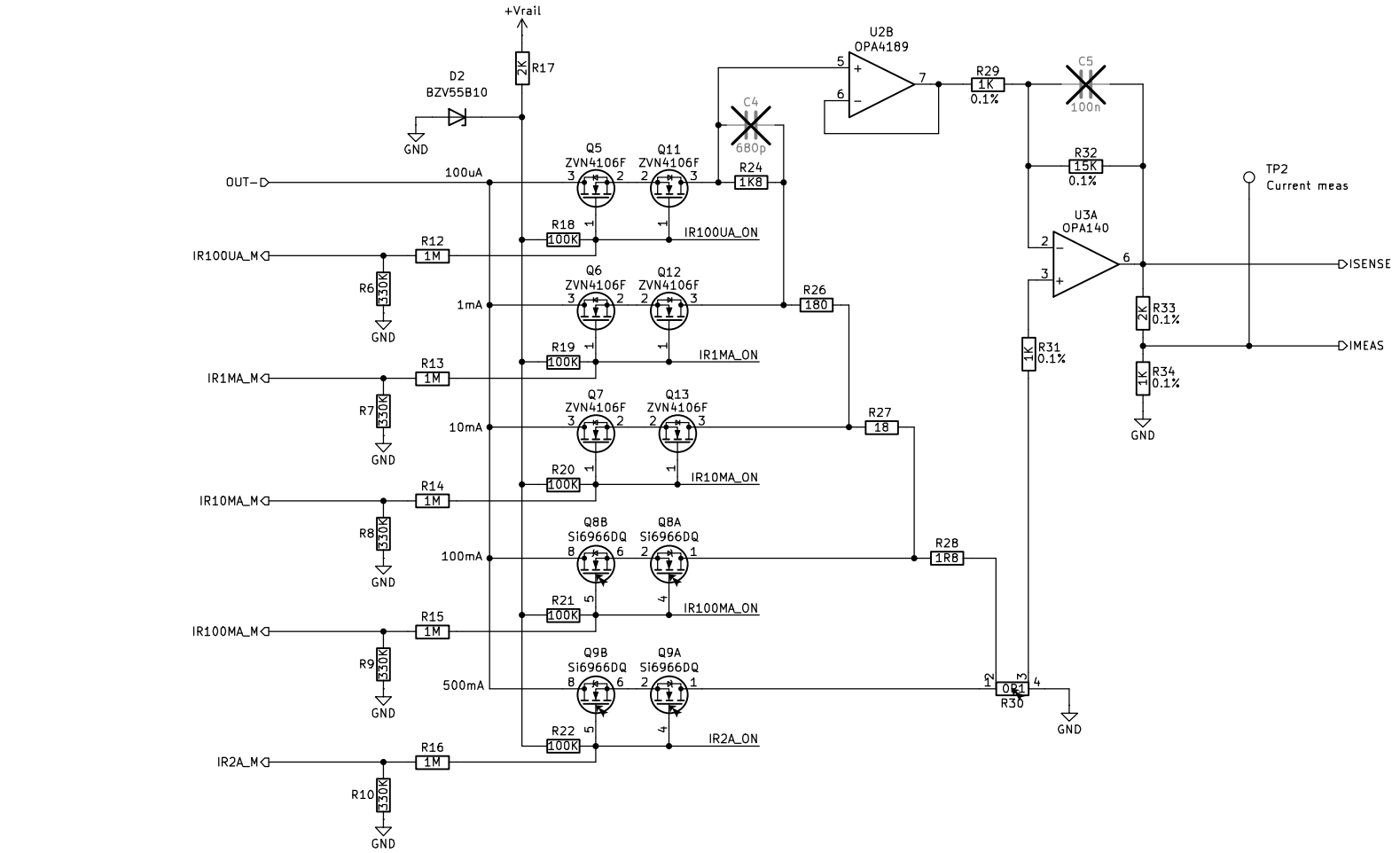

Current Sense
IR2A
IR100MA
IR10MA
IR1MA
IR100UA
IR2A_M
IR100MA_M
IR10MA_M
IR1MA_M
IR100UA_M
OUT−
ISENSED
IMEAS
File: i−sense.kicad_sch
Voltage Sense
MV10M
VRLO
SENSE+
SENSE−
VSENSED
VMEAS
File: v−sense.kicad_sch
ADC & DAC
IMEAS
SCK_ADC
MOSI_ADC
MISO_ADC
CS_ADC
RESET_ADC
DRDY_ADC
SCK_DAC
MOSI_DAC
MISO_DAC
CS_DAC
RESET_DAC
VMEAS
PAOUT
FV
FI
CLAMP+
CLAMP−
File: adc−dac.kicad_sch
Floating MCU
IR2AD
IR100MAD
IR10MAD
IR1MAD
IR100UAD
IR2A_MD
IR100MA_MD
IR10MA_MD
IR1MA_MD
IR100UA_MD
MV10MD
VRLOD
I_VCC
I_GND
I_TX
I_RX
I_CLAMP
OED
FVMODE
File: mcu.kicad_sch
Isolated control port
J3
Power Amplifier
ISENSE
VSENSE
PAOUT
FV
FI
CLAMP+
CLAMP−
FVMODE
File: pa.kicad_sch
K1
DS2−S
+Vrail
RE_OE
OUT+
Output connector
J2
GND
RE_OE
+Vrail
R3
330
D1
BAV70
R1
2K2
Q1
BC847
R2
100K
GND
PWR_FLAG
J1
C1
100u
C2
100u
U1
uA7805
IN
OUT
GND
+5V
C3
22u
−Vrail
TP1
HS1
H1
H2
H3
H4
SMUK
bodri@bodrico.com
Sheet: /
File: sense.kicad_sch
Title: SMUK (SMU by Kapuki)
Size: A4
Date: 2026−10−28
Rev: rev. 1.
KiCad E.D.A. 10.0.4
Id: 1/6

Current conditioning and measurement



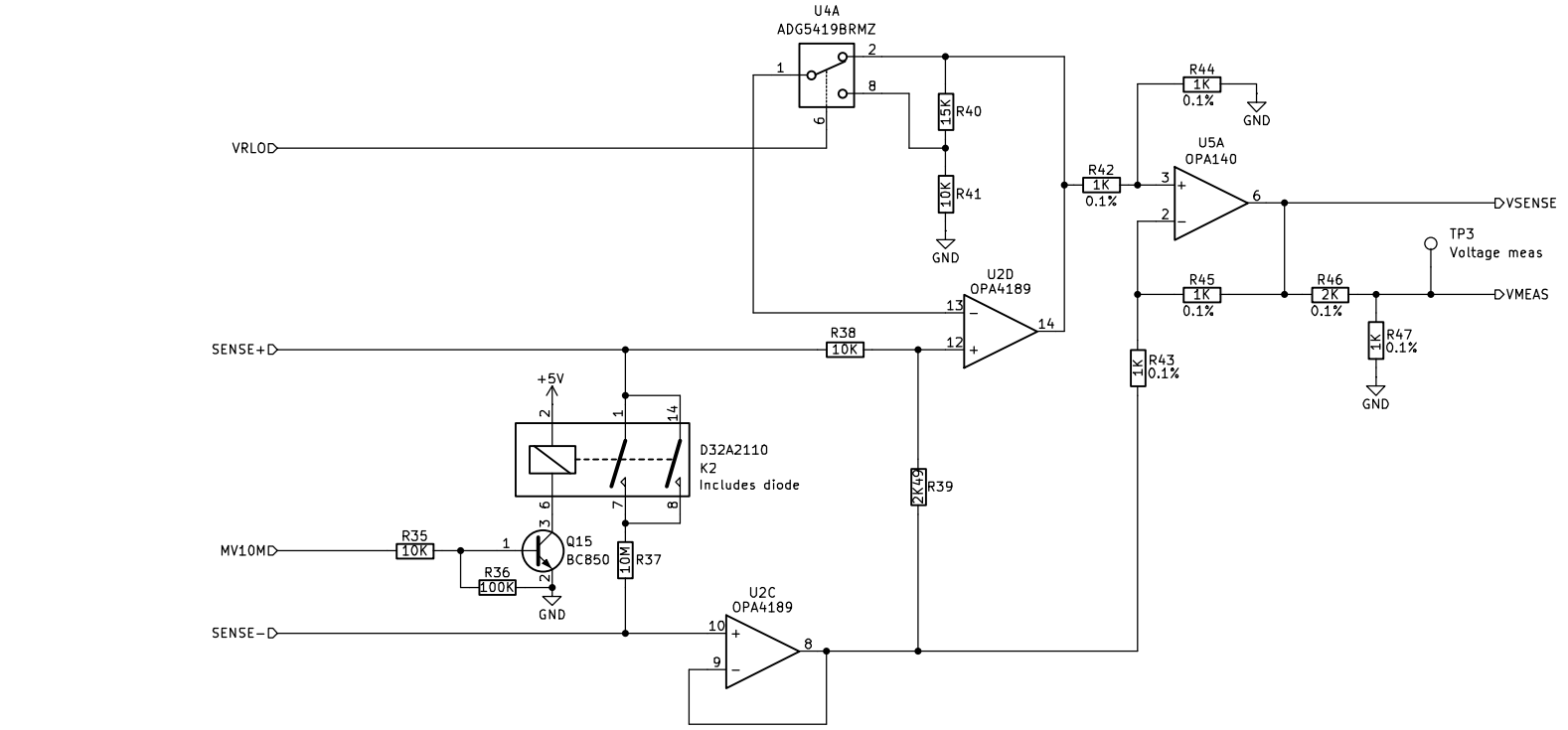
bodri@bodrico.com

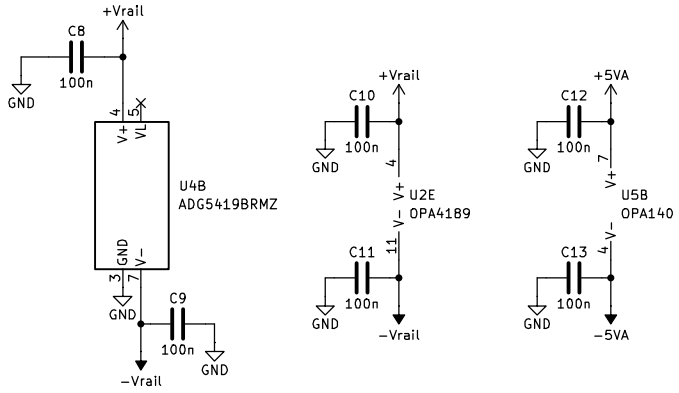
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File: i−sense.kicad_sch

Title: SMUK (SMU by Kapuki)

Size: A4	Date: 2026−10−28	Rev: rev. 1.
KiCad E.D.A. 10.0.4		Id: 2/6

Voltage conditioning and measurement





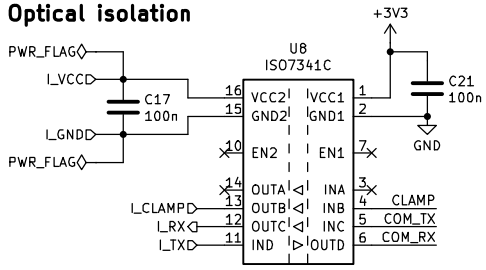
bodri@bodrico.com

Sheet: /Voltage Sense/
File: v−sense.kicad_sch

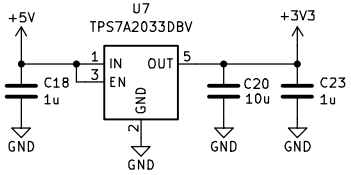
Title: SMUK (SMU by Kapuki)

Size: A4	Date: 2026−10−28	Rev: rev. 1.
KiCad E.D.A. 10.0.4		Id: 3/6

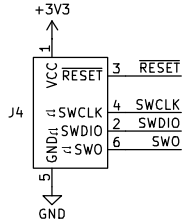
Optical isolation



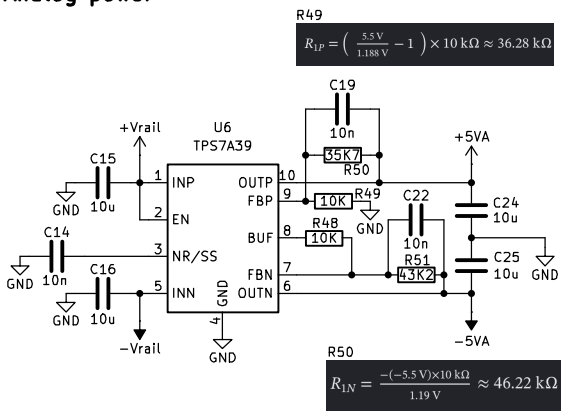
Digital power



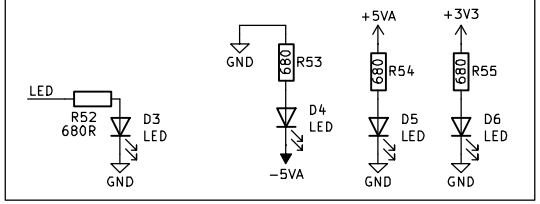
SWD programming header

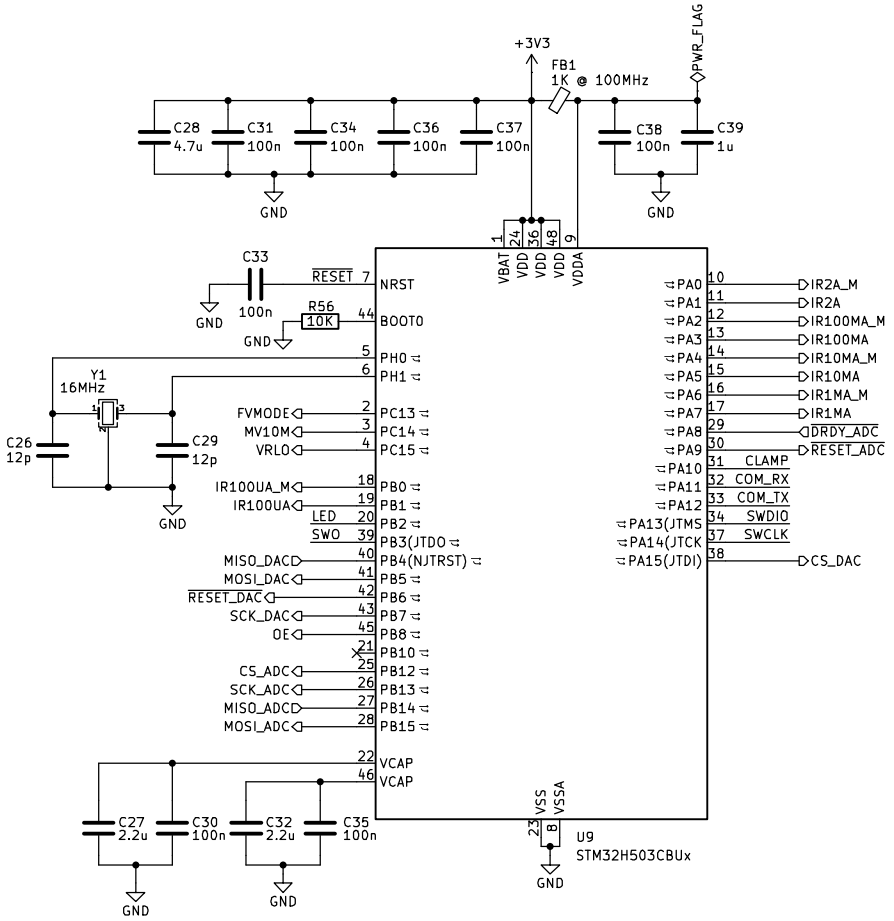


Analog power



Visual feedback





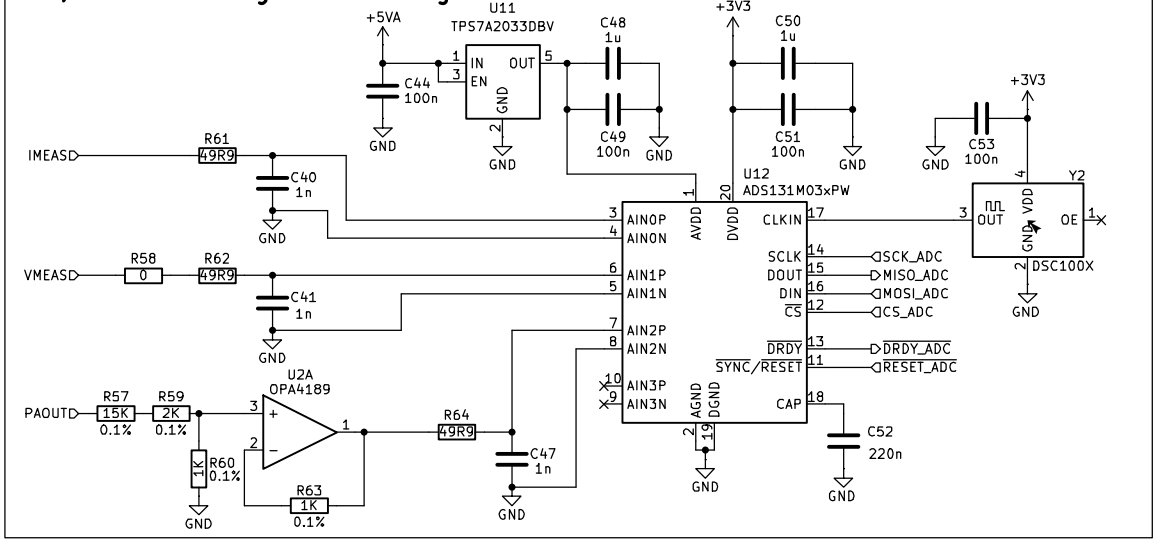
bodri@bodrico.com

Sheet: /Floating MCU/
File: mcu.kicad_sch

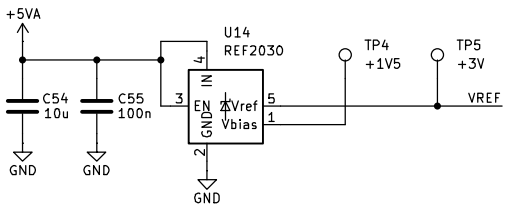
Title: SMUK (SMU by Kapuki)

Size: A4	Date: 2026-10-28	Rev: rev. 1.
KiCad E.D.A. 10.0.4		Id: 4/6

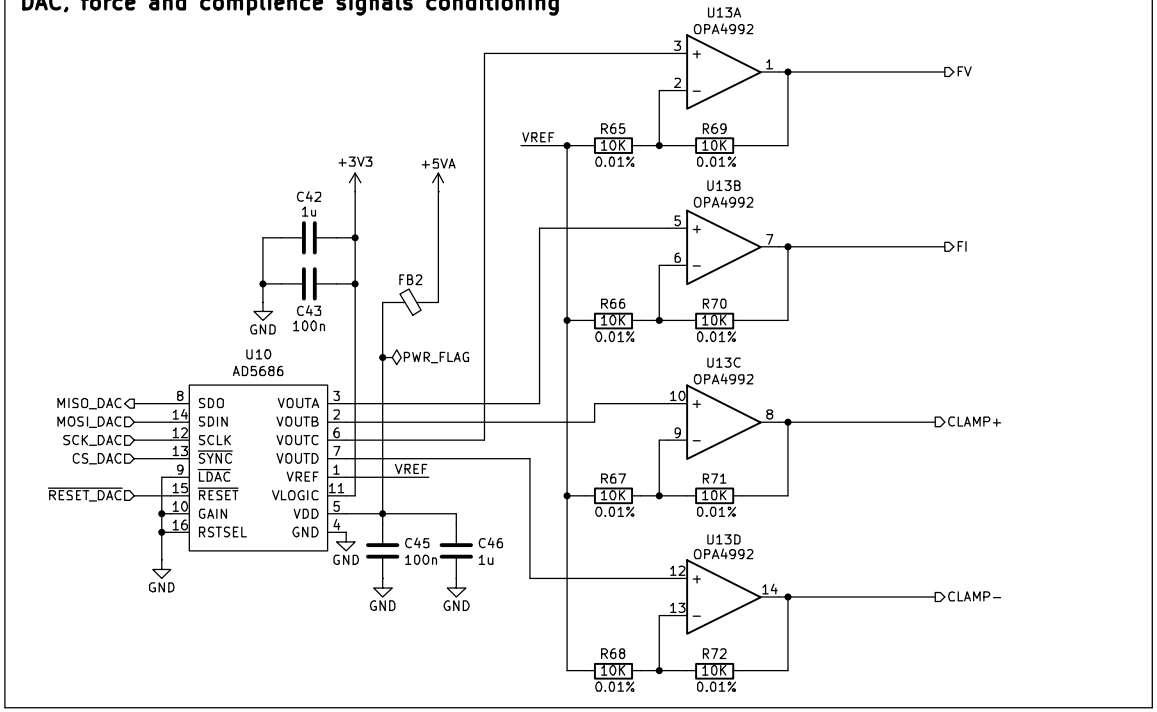
ADC, measurement signal conditioning

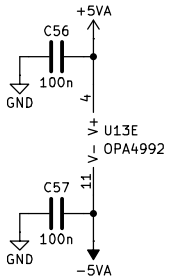


Voltage Reference



DAC, force and complience signals conditioning





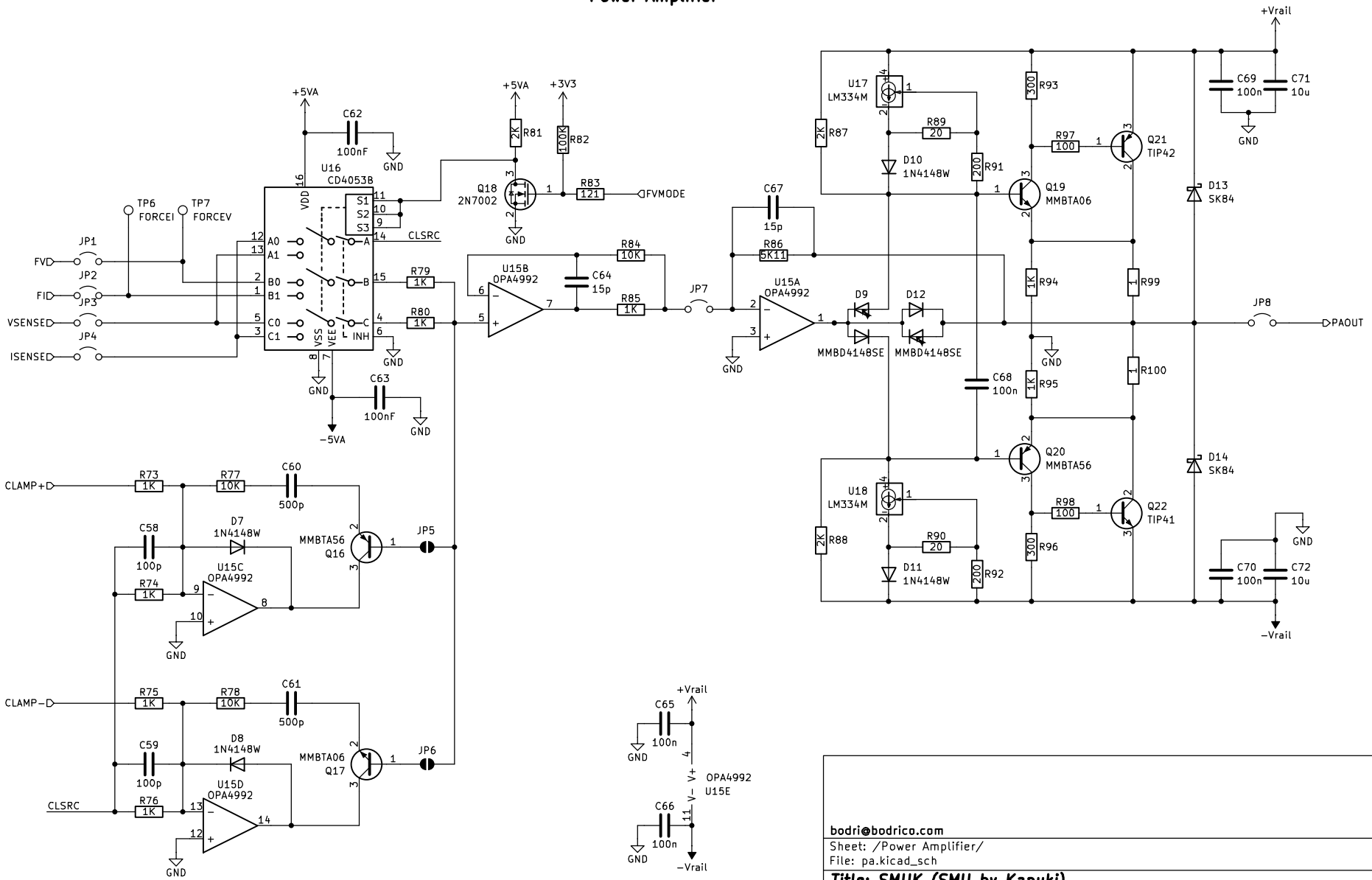
bodri@bodrico.com

Sheet: /ADC & DAC/
File: adc-dac.kicad_sch

Title: SMUK (SMU by Kapuki)

Size: A4	Date: 2026-10-28	Rev: rev. 1.
KiCad E.D.A. 10.0.4		Id: 5/6

Power Amplifier



bodri@bodrico.com

Sheet: /Power Amplifier/
File: pa.kicad_sch

Title: SMUK (SMU by Kapuki)

Size: A4	Date: 2026-10-28	Rev: rev. 1.
KiCad E.D.A. 10.0.4		Id: 6/6